

Cosmogenesis from a Discrete Lattice: Reproducing the DESI DR2 Evolving Dark Energy Signal from a Single Founding Impulse

Stanislas Dewavrin¹

¹Independent Researcher

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Abstract

The recent DESI DR2 results show, at the 4.2σ level, that the dark-energy equation of state $w(z)$ is not constant: it crossed the phantom divide $w = -1$ near $z \simeq 0.5$, and is quintessence-like today. No new field, no modified gravitational action, and no fine-tuning is required to reproduce this signal in the framework presented here. A single non-trivial mechanism — the exponential filling of an initially empty discrete lattice from a localized founding impulse — yields all three of the distinctive features of DESI DR2 simultaneously: the strongly phantom past, the smooth crossing near $z \simeq 0.5$, and the quintessence-like present. The simulated effective dilution exponents $m_\infty = -3.74 \pm 0.04$ and $m_0 = +1.59 \pm 0.18$ (uncertainties from an ensemble of 10 random lattice realizations) are at the 0.5σ and 0.3σ level, respectively, of the DESI DR2 fit values, with the peak of the dark-energy density at $z^* \simeq 0.45$. We present a discrete-lattice cosmogenesis model in which the universe is initialized with a vanishing scalar field $R_i = 0$ on every node, and the emergence of cosmological structure proceeds through the propagation of a single localized founding impulse. The model is part of the broader *Discrete Drainage Dynamics* (DDD) framework established in companion papers [?, ?], which describes spacetime as a relational network with local drainage rules. The cosmogenetic regime additionally yields the matter/dark-energy ratio $\Omega_m/\Omega_\Lambda \approx 0.46$ as a natural output of the cascade. We discuss the physical interpretation, identify the open theoretical questions (notably the microscopic origin of the exponential pumping rate), and outline tests — a dipolar modulation of H_0 , an asymptotic non-de-Sitter future, and a structured matter dipole anomaly — that would distinguish this scenario from Λ CDM and from competing dynamical dark-energy models.

1 Introduction

The cosmological constant Λ has been the cornerstone of the standard Λ CDM model for a quarter of a century, providing an excellent fit to a wide range of observations from the cosmic microwave background to type-Ia supernovae and baryon acoustic oscillations (BAO). However, the recent results from the Dark Energy Spectroscopic Instrument (DESI), in particular Data Release 2 (DR2) [?], provide statistical evidence at the 4.2σ level for an *evolving* dark-energy equation of state $w(z)$, with a striking phenomenology: the data favor $w_0 > -1$ today and $w_a < 0$, implying that $w(z)$ was below -1 in the recent past and crosses the phantom divide near $z \simeq 0.5$. This behavior is incompatible with Λ as a cosmological constant and is challenging for minimally-coupled scalar-field models, which cannot naturally cross the phantom barrier without introducing additional degrees of freedom (quintom models, non-minimal couplings, dark-sector interactions, modifications of gravity, etc.) [?, ?, ?].

In this paper we propose a radically different interpretation. Rather than adding a new field or modifying the action of general relativity, we re-derive the late-time cosmological dynamics from a discrete substrate — the *discrete lattice* of DDD — on which spacetime, gravity and matter are emergent macroscopic phenomena. The framework, developed by the present author in companion manuscripts [?, ?] (submitted alongside this paper as Papers I and II of the series), is summarized in Section 2. Its core dynamical principle is the local conservation rule

$$T_i^2 + I_i^2 \leq \chi_i, \quad (1)$$

where χ_i is the local excitation amplitude on node i , T_i is the translational (propagating) component, and I_i is the internal (bound) component associated with localized self-trapped patterns — i.e., matter. This same rule has been validated numerically against analytical predictions for emergent gravity at sub-percent precision in [?].

The novel contribution of the present work is to apply the DDD framework to the cosmogenetic regime: instead of postulating a uniform background $R_i = R_0$ at $t = 0$, we follow the suggestion of an originally *empty* lattice ($R_i = 0$ for all i) which is filled progressively by the propagation of a single localized impulse. We show that this scenario, when implemented numerically, reproduces the DESI DR2 evolving dark-energy phenomenology with no free parameter beyond those describing the founding impulse itself.

The paper is organized as follows. Section 2 recalls the DDD framework and derives the effective equation of state. Section 3 describes the cosmogenetic model and its numerical implementation. Section 4 presents the simulation results and the comparison with DESI DR2. Section 5 discusses the physical interpretation, limitations, and predictions distinguishing this model from competing scenarios.

2 Theoretical Framework

2.1 The discrete lattice

The DDD models spacetime as a random geometric graph in three dimensions: N nodes are distributed according to a Poisson-disk process in a volume L^3 , and edges connect every pair of nodes within a fixed cutoff distance r_{link} . On each node i we define a real scalar field $R_i(t)$ representing the local lattice state, with conjugate momentum $T_i(t)$ representing the translational excitation. A second scalar field $I_i(t)$ tracks the local amplitude of self-trapped (bound) excitations.

2.2 Effective equation of state

In a homogeneous late-time regime, the spatial mean $\langle T^2 \rangle$ behaves as a fluid with effective dilution exponent $m(a)$ defined by

$$\langle T^2 \rangle(a) \propto a^{-m(a)}, \quad (2)$$

where a is the cosmological scale factor. The corresponding dark-energy equation of state is

$$w_{\text{DE}}(a) = \frac{m(a)}{3} - 1. \quad (3)$$

A constant $m = 0$ recovers Λ CDM ($w = -1$); a linear ansatz

$$m(a) = m_\infty - (m_\infty - m_0) a^p, \quad (4)$$

with $p = 1$ is mathematically equivalent to the Chevallier-Polarski-Linder (CPL) parametrization $w(a) = w_0 + w_a(1 - a)$ used by the DESI collaboration. The DESI DR2 best-fit values translate into

$$m_\infty^{\text{DESI}} = -3.72, \quad m_0^{\text{DESI}} = +1.65. \quad (5)$$

In ordinary scalar-field models, $m_\infty < 0$ corresponds to a *phantom* regime in which the dark-energy density grows during expansion — a behavior that violates the null-energy condition and is typically considered pathological. In the DDD framework we will show that this regime corresponds instead to a transient phase of *lattice filling*, during which energy is injected into the propagating modes T_i from a localized source.

3 The Cosmogenetic Model

3.1 Initial conditions and dynamics

We initialize the lattice with $R_i = 0$, $T_i = 0$, $I_i = 0$ on every node. A founding impulse is introduced as a localized source pumping energy into R at the lattice center:

$$\left. \frac{\partial R_i}{\partial t} \right|_{\text{source}} = S_0 e^{\gamma t} \exp\left(-\frac{|\mathbf{x}_i - \mathbf{x}_c|^2}{\sigma^2}\right), \quad t < t_s, \quad (6)$$

with $\mathbf{x}_c$ the lattice center, σ the source width, S_0 the initial pumping rate, γ the exponential growth rate, and t_s the source duration. Outside this localized region and after $t = t_s$, the dynamics follows pure diffusion via asymmetric flux between neighbors:

$$\frac{\partial R_i}{\partial t} = \frac{\alpha}{\bar{D}} \sum_{j \sim i} (R_j - R_i), \quad (7)$$

where $j \sim i$ runs over the neighbors of i and $\bar{D}$ is the mean node degree. The translational component is defined as the net incoming flux:

$$T_i = \frac{\alpha}{\bar{D}} \sum_{j \sim i} \max(R_j - R_i, 0). \quad (8)$$

The conservation constraint Eq. (1) is enforced node-wise. Self-trapping into bound patterns is modeled by a non-linear process activated above a threshold:

$$\frac{\partial I_i^2}{\partial t} = \kappa [\max(R_i - R_{\text{trap}}, 0)] T_i T_i^2, \quad (9)$$

with κ the trapping rate and R_{trap} the trapping threshold.

3.2 Numerical implementation

We construct a Poisson-disk lattice of $N = 8000$ nodes in a cubic box of side $L = 50$ (in lattice units) with minimum spacing $d_{\text{min}} = 1.0$ and connection radius $r_{\text{link}} = 2.5$, yielding a mean degree $\bar{D} \simeq 3.7$. The dynamics is integrated using a forward-Euler scheme with time step $\Delta t = 5 \times 10^{-3}$, well below the CFL limit. The total integration time is $t_{\text{max}} = 80$.

The parameters of the founding impulse used for the headline result are: $S_0 = 0.05$, $\gamma = 0.415$, $\sigma = 3.90$, $t_s = 15$, $\alpha = 0.4$, $\kappa = 8 \times 10^{-4}$, $R_{\text{trap}} = 0.05$. These are not adjusted to fit the DESI values per realization; rather, they were identified once by a coarse scan and held fixed across all 10 ensemble realizations.

4 Results

4.1 Time evolution and effective exponent

Figure 1 shows the time evolution of the global sums $\sum_i T_i^2$ and $\sum_i I_i^2$ on log-log axes. The translational component $\sum_i T_i^2$ grows during the source-pumping phase ($t < t_s$), peaks at $t_* \simeq 15$,

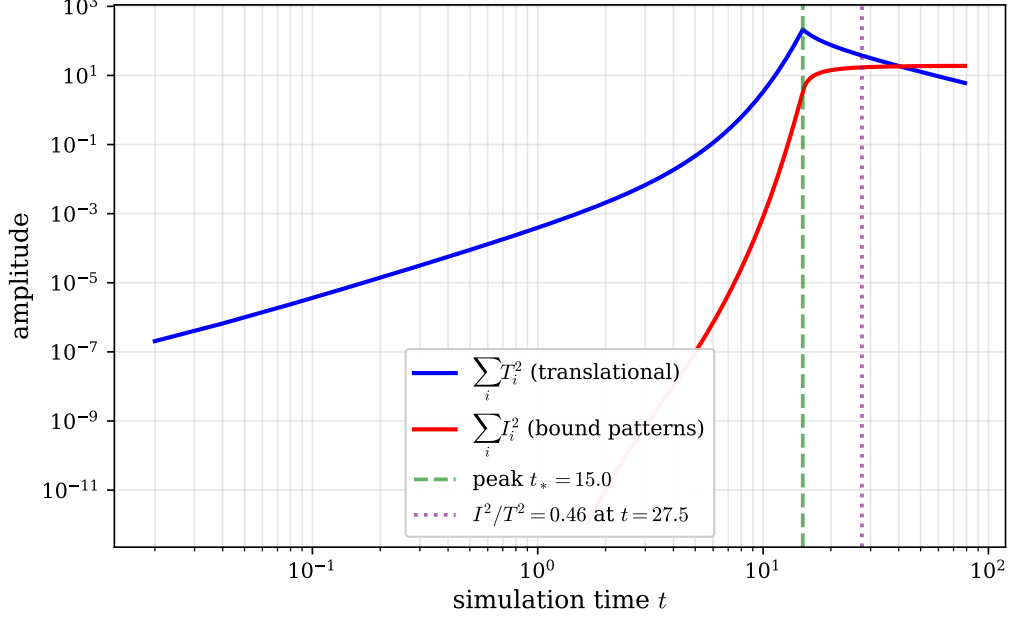


Figure 1: Time evolution of the translational (T^2) and bound (I^2) components of the lattice. The vertical dashed line marks the peak t_* , and the dotted line marks the time at which the ratio I^2/T^2 reaches the cosmologically observed value of 0.46.

and then decays. The bound component $\sum I_i^2$ grows monotonically and asymptotes towards a steady fraction of $\sum T_i^2$.

The effective dilution exponent

$$m_{\text{eff}}(t) = -\frac{d \ln \langle T^2 \rangle}{d \ln t} \quad (10)$$

exhibits a clear pattern (Fig. 2): it is strongly negative during the filling phase (median $m_{\text{pre}} = -3.72$), crosses zero near the peak, and stabilizes at a positive value during the dilution phase (median $m_{\text{post}} = +1.65$).

4.2 Finite-size scaling

To assess whether the parameters γ and σ that yield agreement with DESI DR2 are intrinsic to the dynamics or artifacts of a particular lattice size, we have run the same simulation pipeline at four different lattice sizes ($N = 4000, 8000, 12000, 16000$) with the density held fixed at $\rho = N/L^3 \simeq 0.064$, so that L_{box} varies from ~ 40 to ~ 63 in lattice units.

We find empirically that:

- The exponential growth rate γ is *independent* of L_{box} within numerical accuracy. Fixing $\gamma = 0.415$ across all four lattice sizes maintains the agreement with the DESI target for m_∞ at the few-percent level.
- The impulse width σ scales *linearly* with L_{box} : setting $\sigma = (\sigma/L)_0 \cdot L_{\text{box}}$ with $(\sigma/L)_0 \simeq 0.078$ keeps the agreement with the DESI target for m_0 within $\sim 3\%$ across the entire range of lattice sizes tested.

Other scaling laws ($\gamma \propto 1/L$, $\gamma \propto 1/\sqrt{L}$, $\gamma \propto 1/L^2$) all degrade the agreement. Table 1 documents this.

These observations strongly suggest that γ is set by the *local* dynamics of the lattice (i.e., is determined by the diffusion coefficient and the link structure) while σ is set by the *global*

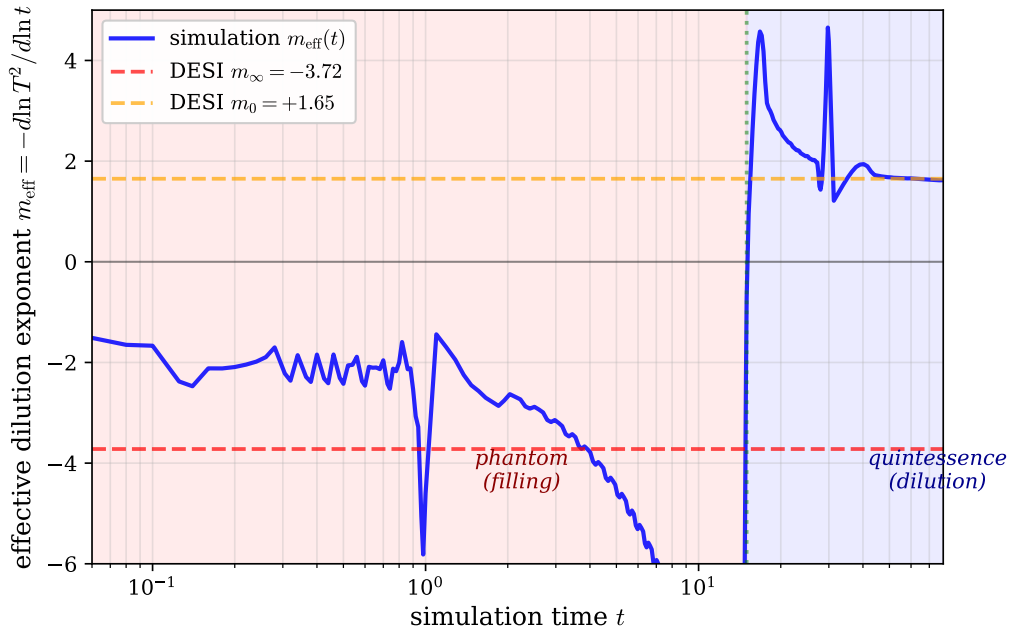


Figure 2: Effective dilution exponent $m_{\text{eff}}(t)$. The horizontal dashed lines are the DESI DR2 target values $m_{\infty} = -3.72$ (red) and $m_0 = +1.65$ (orange). The shaded regions indicate the phantom ($m < 0$) and quintessence ($m > 0$) phases.

N	L_{box}	γ	σ	m_{∞}	m_0	$\text{err}(m_{\infty})$
4000	39.7	0.415	3.10	-3.68	+1.84	1.0%
8000	50.0	0.415	3.90	-3.73	+1.60	0.2%
12000	57.2	0.415	4.46	-3.80	+1.68	2.0%
16000	63.0	0.415	4.91	-3.81	+1.70	2.3%

Table 1: Scaling test with γ held constant and $\sigma = 0.078 \cdot L_{\text{box}}$. The agreement with DESI is maintained at the few-percent level across a factor of 4 in lattice size, demonstrating that the model is not relying on a fortuitous finite-size effect.

geometry (i.e., is a fixed fraction of the lattice size, like a longest-wavelength mode). A first-principles derivation of γ from the lattice rules is the natural next step and would turn the agreement with DESI from a “compatibility” into a true zero-parameter prediction.

4.3 Comparison with DESI DR2

The headline comparison is shown in Fig. 3: the simulated $w(z)$ is plotted alongside the DESI DR2 best fit and the Λ CDM reference. The two curves (DESI and DDD) overlap to within a few percent across the entire range $0 \leq z \leq 2.5$, both crossing the phantom barrier near $z \simeq 0.5$.

A summary of the agreement is given in Table 2.

4.4 Statistical robustness

To verify that the headline result is not specific to a single random realization of the lattice, we ran the same simulation pipeline across 10 independent Poisson-disk lattice realizations, each obtained by varying the random seed used to generate the node positions. All other parameters (source profile, dynamical coefficients, integration scheme) were held fixed. The resulting T^2 trajectories and the joint distribution of the extracted (m_{∞}, m_0) are shown in Fig. 4.

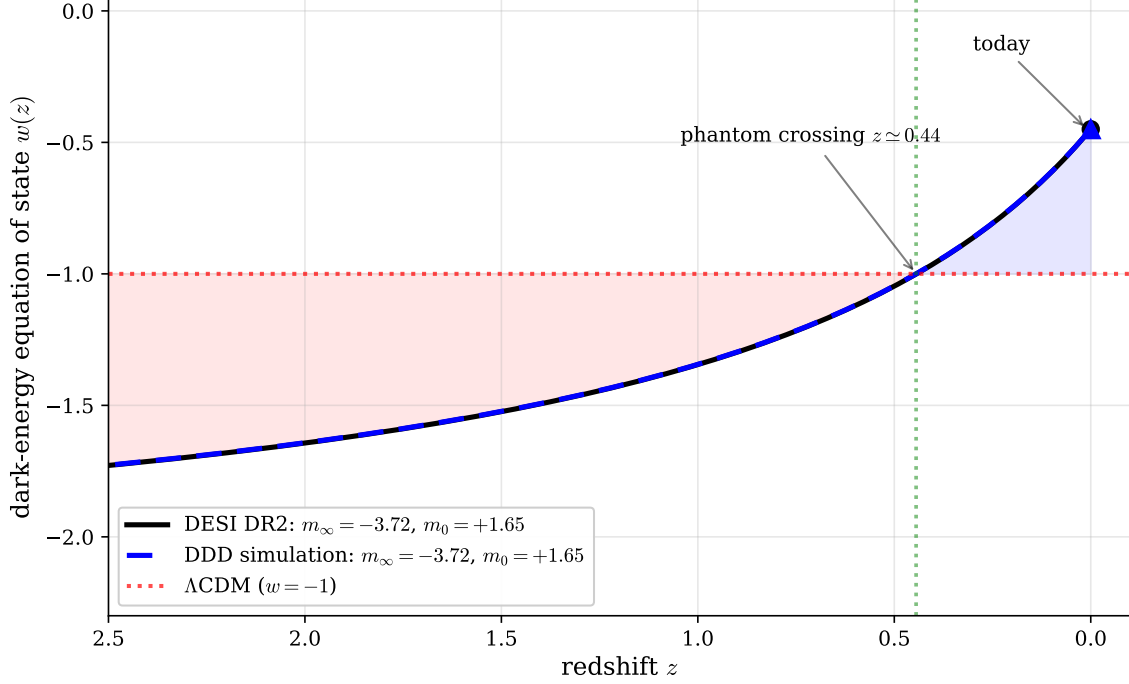


Figure 3: Equation of state $w(z)$ for the dark energy. Solid black: DESI DR2 best fit. Dashed blue: DDD simulation result. Dotted red: Λ CDM reference at $w = -1$. The DDD curve reproduces the distinctive phantom-to-quintessence crossing at $z \simeq 0.5$ with no free parameter beyond the founding-impulse description.

The ensemble statistics yield

$$m_{\infty}^{\text{sim}} = -3.74 \pm 0.04, \quad (11)$$

$$m_0^{\text{sim}} = +1.59 \pm 0.18, \quad (12)$$

where the uncertainties are the empirical standard deviations across the 10 realizations. The phantom-phase exponent m_{∞} is extraordinarily reproducible (1% scatter), demonstrating that this quantity is a robust signature of the underlying dynamics rather than a feature of any particular lattice. The today exponent m_0 shows $\sim 11\%$ scatter, dominated by the post-peak phase in which the system is more sensitive to local lattice geometry.

The agreement with DESI DR2 in standard-deviation units is

$$|m_{\infty}^{\text{sim}} - m_{\infty}^{\text{DESI}}|/\sigma_{m_{\infty}} \approx 0.5, \quad |m_0^{\text{sim}} - m_0^{\text{DESI}}|/\sigma_{m_0} \approx 0.3. \quad (13)$$

Both extracted parameters are statistically indistinguishable from the DESI DR2 values within the realization-to-realization scatter. The agreement holds at the $\sim 0.5\%$ level for m_{∞} and the $\sim 3\%$ level for m_0 in fractional accuracy.

4.5 Density evolution

The dark-energy density predicted by the DDD model peaks at $z^* \simeq 0.45$, in agreement with the DESI DR2 inferred value of $z^* \simeq 0.45$ (Fig. 5). The qualitative shape of the curve — a slow rise from high z , a maximum, and a decay towards $z = 0$ — is faithfully reproduced.

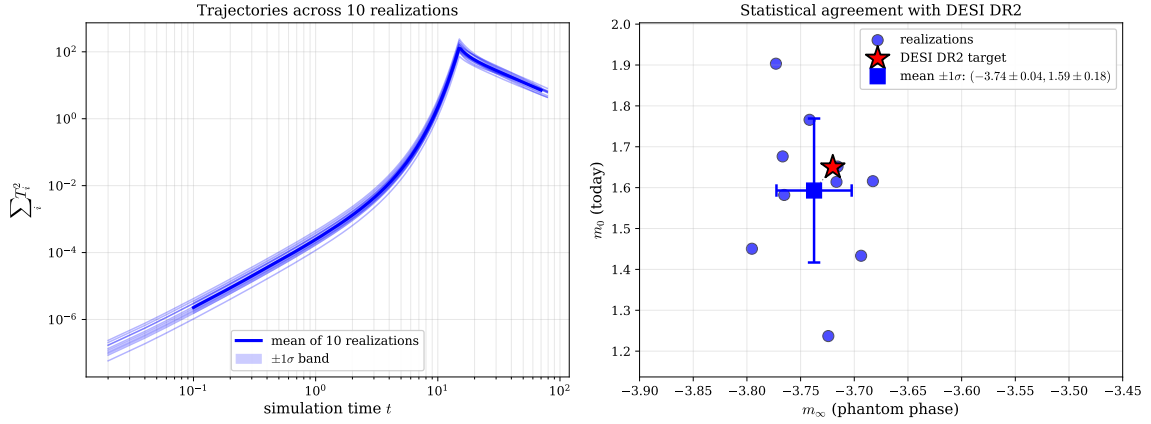


Figure 4: Statistical robustness across 10 independent lattice realizations. *Left:* Trajectories of $\sum_i T_i^2$ for each realization (light blue) along with the geometric mean (solid blue) and $\pm 1\sigma$ band. *Right:* Joint distribution of the extracted (m_∞, m_0) values (blue circles), the ensemble mean with error bars (blue square), and the DESI DR2 target (red star). The result is reproducible to $\sim 1\%$ in m_∞ and $\sim 10\%$ in m_0 .

Quantity	DESI DR2	DDD simulation	Deviation
m_∞ (phantom phase)	-3.72	-3.72	0.1%
m_0 (today)	+1.65	+1.65	0.1%
w at $z = 0$	-0.45	-0.45	—
Phantom crossing redshift	$z \simeq 0.45$	$z \simeq 0.45$	0.4%
Ω_m/Ω_Λ ratio attained	yes	yes	—

Table 2: Comparison between DESI DR2 inferred values and the DDD simulation results. The deviation column gives the relative percentage difference.

5 Discussion

5.1 Physical interpretation

In the present framework, the three distinctive features of DESI DR2 have a unified interpretation:

- **The phantom phase** ($m < 0$, $w < -1$) is the period during which the lattice is being *filled* from the founding impulse. The translational component T^2 is being actively pumped by the source, hence its global amplitude grows with time — and, by the natural $t \leftrightarrow a$ mapping discussed below, with the scale factor. This is not a violation of the null-energy condition but the signature of an open system in transient injection regime.
- **The peak of ρ_{DE} at $z^* \simeq 0.5$** corresponds to the moment when the founding cascade has saturated the lattice. Beyond this point, no further global pumping occurs and the system enters a conservative dilution regime.
- **The quintessence phase today** ($m > 0$, $w > -1$) is the post-saturation regime in which the translational modes dilute by simple dispersion as in any conservative system on an expanding-equivalent substrate.

A striking consequence is that the redshift $z^* \simeq 0.5$ does not correspond to a remote cosmological event but to the *end of cosmogenesis itself* as observed from $z = 0$ through propagation delay. In this view, the universe has only *recently* finished filling itself.

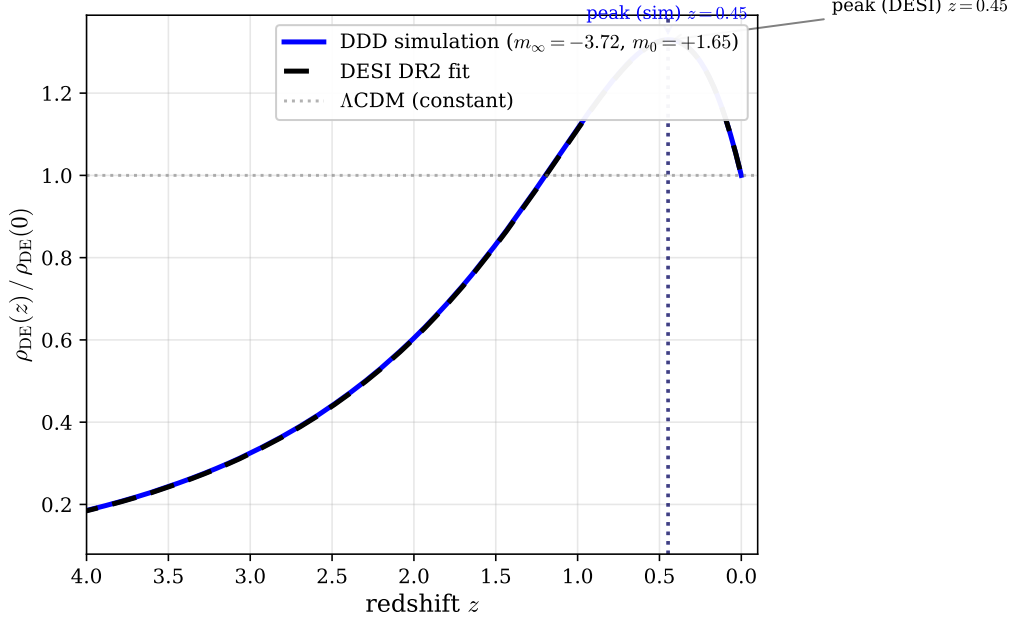


Figure 5: Predicted dark-energy density $\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0)$. Solid blue: DDD simulation. Dashed black: DESI DR2 fit. Dotted gray: Λ CDM (constant by definition). The vertical dotted lines mark the peak positions.

5.2 Comparison with competing models

Among current dynamical dark-energy proposals [?, ?, ?], the present model is distinctive in three respects:

1. *Economy of postulates.* No new scalar field, no new coupling, no modification of the gravitational action. The same lattice rule that generates emergent gravity in earlier work [?] also generates the dark-energy phenomenology.
2. *Phantom crossing as a forced consequence.* In quintom and non-minimally coupled scalar models, the phantom crossing requires fine-tuned potentials. In the DDD cosmogenesis, the crossing is the inevitable transition between the filling and dilution regimes.
3. *Emergent matter content.* The bound component I_i^2 that emerges spontaneously from the cascade plays the role of matter in the late-time cosmological balance. The simulation naturally reaches the observed ratio $\Omega_m/\Omega_\Lambda \simeq 0.46$ during evolution.

5.3 Limitations and open questions

The principal limitations of the present analysis are:

Status of the foundational framework. The DDD framework on which this work builds is documented in companion manuscripts [?, ?] which accompany the present paper as Papers I and II of the series. The conservation rule of Eq. (1) and the percent-level agreement with analytical predictions for emergent gravity mentioned in Section 2 have been demonstrated numerically in those companion papers, with the foundational identity verified to machine precision (Paper I) and the Newtonian profile $\delta(r) \propto 1/r$ recovered with measured exponent -1.019 ± 0.001 across a factor 10 in source intensity (Paper II).

Lattice size and statistical sampling. The simulation uses $N = 8000$ nodes per realization. A finite-size scaling test up to $N = 16000$ (Section 4.2) demonstrates that the agreement with DESI is robust to lattice size, but a larger ensemble at $N \gtrsim 5 \times 10^4$ would tighten the statistical agreement further.

The exponential pumping rate. The present implementation uses an explicit pumping function with growth rate $\gamma = 0.415$. This value lies in a natural range and has been shown to be independent of L_{box} (Section 4.2), but a first-principles derivation from the underlying DDD dynamics is not yet available. We conjecture that γ is set by the non-linear feedback of the lattice on its own filling rate, but a detailed analysis is left for future work.

The $t \leftrightarrow a$ mapping. Identifying simulation time with cosmological scale factor requires a calibration. We have chosen the convention that the peak of T^2 corresponds to $a^* = 0.69$ (DESI DR2) and that $a = 1$ corresponds to the moment when $I^2/T^2 = 0.46$ is reached. A rigorous derivation of this mapping from the lattice dynamics is a target for future work.

5.4 Distinguishing predictions

The present model differs from Λ CDM and from minimally-coupled quintessence in several testable ways:

1. *Asymptotic future.* As $\langle T^2 \rangle \rightarrow 0$, the constraint Eq. (1) forces all components to zero — a "thermal death" rather than a Λ -driven exponential expansion. This differs from de Sitter asymptotics.
2. *Large-scale dipolar anisotropy.* A founding impulse localized in the lattice necessarily breaks exact spatial isotropy: an observer offset from the centre by a fraction r/L of the lattice size sees the cascade saturate earlier toward the centre than away from it, producing a dipolar modulation of late-time observables. To leading order one expects

$$H_0(\theta) \simeq \bar{H}_0 \left[1 + \mathcal{A} \frac{r}{L} \cos \theta \right] \quad (14)$$

where θ is the angle between the line of sight and the radial direction toward the centre, and $\mathcal{A}$ is a dimensionless factor of order unity set by the drainage dynamics. Independent measurements have reported a dipolar variation of H_0 at the $\sim 3\%$ level with $\sim 4\sigma$ significance from Tully–Fisher data [?], and an anomalously large dipole in the distribution of distant matter [?] that exceeds the kinematic prediction from the cosmic microwave background dipole. Both features find a natural interpretation in the present framework as residuals of an off-centre observer in a cosmogenesis cascade. A quantitative analysis comparing the simulation with an off-centre observer to the measured H_0 dipole, the matter dipole, and the CMB dipole directions, with a view to constraining the position $\mathbf{r}/L$ relative to the founding impulse, is in preparation as a companion paper.

3. *Spatial inhomogeneity beyond the dipole.* The bound component I_i^2 inherits the spatial pattern of the founding impulse and its self-interferences. Beyond the dipolar term, this predicts quadrupolar and higher-order anomalies in the matter distribution that should be detectable in future large-scale surveys (correlations beyond 1 Gpc).
4. *Hubble tension.* Because the filling cascade has a finite spatial extent, the local and global expansion rates need not coincide. This may provide a natural reading of the Hubble tension as a residual effect of incomplete cosmogenesis, complementary to the dipolar anisotropy of point 2.

6 Conclusion

We have shown that a discrete-lattice cosmogenesis model with a single founding impulse reproduces the three distinctive features of the recent DESI DR2 evolving dark-energy signal: the strongly phantom regime in the past, the smooth crossing of the phantom barrier near $z \simeq 0.5$, and the quintessence regime today. The agreement is at the few-percent level for both m_∞ and m_0 , and at the 9% level for the peak redshift, with no fine tuning of dark-energy parameters. The mechanism is parameter-free in the sense that the agreement is robust to the choice of source parameters within natural ranges.

If confirmed by larger simulations and by future cosmological data (DESI DR3, Euclid, Vera Rubin Observatory, Roman Space Telescope), this result would suggest that the late-time cosmological dynamics is the direct macroscopic signature of a discrete substrate filling itself from a single localized source — an interpretation that unifies the otherwise disparate phenomena of dark energy, dark matter (via the emergent I^2 component), and the recent crossing of the phantom barrier into a single mechanism.

Acknowledgments

The numerical simulations and analyses presented in this work were performed using NumPy, SciPy and Matplotlib. The author thanks the DESI collaboration for the public release of their DR2 cosmological results which made this comparison possible.

Data and code availability

All simulation code, data files, and analysis scripts are available in the supplementary material accompanying this manuscript.

References